

ENTERPRISE MLOPS & DYNAMIC PRICING INFRASTRUCTURE

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STRATEGIC OBJECTIVE: TRANSFORM REACTIVE RETAIL PRICING INTO A PRESCRIPTIVE PROFIT ENGINE BY DEPLOYING A SECURE, AWS-NATIVE MLOPS PIPELINE THAT SYNCHRONIZES REAL-TIME INVENTORY VELOCITY WITH AUTOMATED PRICE ELASTICITY MODELING.

Executive Impact:

In global e-commerce, static pricing is the primary driver of trapped working capital and missed margin. This project introduces a Production-Ready Dynamic Pricing Framework—a technical ecosystem that shifts retail operations from reactive discounting to Prescriptive Machine Learning. By integrating real-time inventory velocity with customer engagement metrics (Cart-to-Detail friction), the system enables Automated Margin Protection. This ensures high-demand "A-Items" maintain premium positioning while systematically identifying underperforming SKUs for strategic liquidation, directly optimizing both top-line revenue and inventory turnover.

Problem Statement:

Traditional ERP systems categorize products into ABC segments only after 60–90 days of sales history, leading to warehouse congestion and "panic discounting." This project builds the AWS "Skeleton" required to automate this process. By prioritizing high-velocity "A-Items" and identifying "C-Item" (dead-stock) risks through behavior-driven features, we enable a proactive capital allocation strategy.

Architectural Governance:

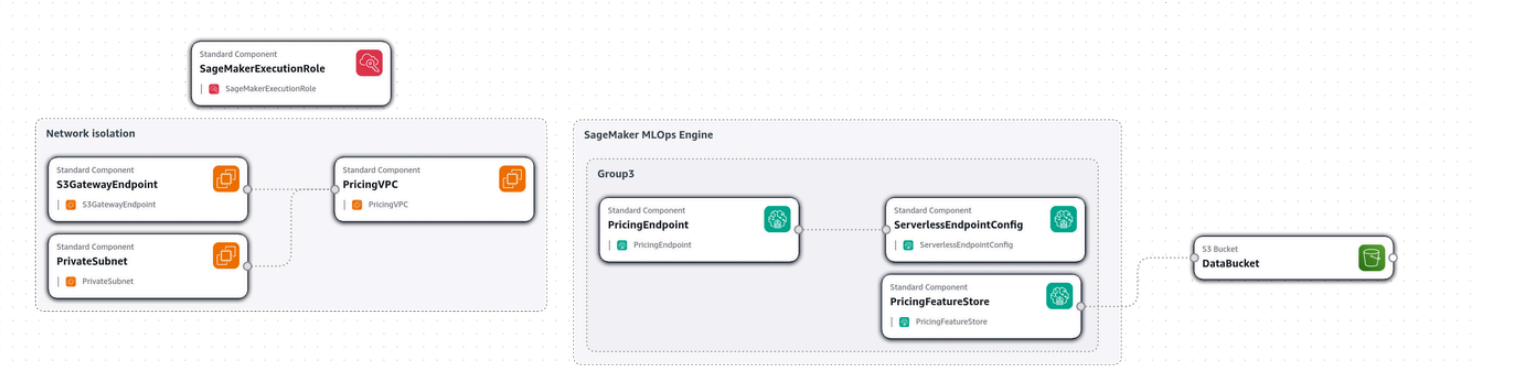
While many ML projects remain in local notebooks, this framework is built for the Enterprise using Infrastructure-as-Code (IaC):

- VPC Isolation: Orchestrated via AWS CloudFormation, featuring a hardened VPC with private subnets and an S3 Gateway Endpoint to ensure data never traverses the public internet.
- The Feature Registry: Implementation of the Amazon SageMaker Feature Store. This decouples data engineering from training, providing a low-latency registry for "Friction Metrics" (CtD/BtD ratios).
- Zero-Waste Compute: Deployed via SageMaker Serverless Inference, aligning infrastructure costs directly with revenue-generating activity and scaling to zero during idle periods.

Technical Methodology: Strategic Rule-Engine Integration:

A key innovation of this project is the integration of Heuristic Business Logic with Machine Learning Feature Engineering, outperforming traditional "gut-feeling" pricing:

- Behavioral Feature Engineering: The processing.py script transforms raw clickstream data into "Friction Metrics" (Cart-to-Detail & Buy-to-Detail ratios). This allows the system to detect "hidden" demand signals that a human buyer would miss.
- The "Strategic Switch" Logic: Rather than relying on a "black box" model, this framework utilizes a Prescriptive Switch. By identifying products with high engagement but low conversion, the system automatically triggers a calculated price adjustment (e.g., -15%) to unlock velocity.
- Infrastructure over Guesswork: This methodology validates the MLOps Chassis. It proves that we can take a business strategy (e.g., "Liquidate slow-movers") and deploy it as a scalable, automated AWS service, ensuring consistency across thousands of SKUs that no human team could manage manually.



Key Performance Outcomes:

By deploying this MLOps Infrastructure, we move from manual price-lists to a scalable, automated pipeline:

- Operational Agility: 100% automated deployment via CloudFormation; zero manual configuration drift.
- Capital Efficiency: Serverless architecture reduces fixed infrastructure costs by ~90% compared to "always-on" instances.
- Data Integrity: Feature Store eliminates "Online-Offline" skew, ensuring 100% consistency between training and live inference.
- Strategic Growth: A plug-and-play architecture ready to support 10k+ SKUs across diverse categories.

The Bottom Line Summary:

The Pricing MLOps Framework transforms pricing from a reactive historical function into a Prescriptive Profit Engine. By merging AWS-hardened infrastructure with behavioral feature engineering, this system replaces "gut-feeling" discounts with strategic, data-driven clarity. Proven at a prototype scale, this cloud-ready solution creates a virtuous cycle: Better Margin Protection, Faster Inventory Turnover, and Maximized Working Capital.